

The movie was fantastic and thrilling!  
Label: positive

I hated the film; it was boring and slow.  
Label: negative

What a masterpiece, truly inspiring!  
Label: positive

The plot was dull and characters uninspiring.  
Label: negative

{Evaluation sample}  
Label: **positive/negative**

Figure 6. Few-shot evaluation prompts. **text** indicates the labels predicted by the model.

- The suspense in the final act left room for the most important and most thrilling of all parts of the movie
- The new 'Innate Robots' is a must-see for anyone who loves the latest in the field technology
- The critically acclaimed action movie "Fast and Far East City" is a work of the highest caliber ever to be made
- The action-movie 'Ace Driver' is a 'wins' masterpiece that will leave you feeling 'th

#### C.2.2. NEGATIVE LABEL

- The action level plot of the movie was not up to mark. The use cases were not engaging and the the use of the provided
- The movie was terrible. You are writing a one page story set in a world where people can only see the world
- The new movie was absolutely not enjoyable. The over-dilting of the water made it a real downer experience
- The action-adventure filled movie was a disappointment. The excessive use of the 't' sound
- he quality of the action in Bad Movie is not up to the standard set by the other 'g' and movie.

### C.3. Synthetic Tweet Emotions Samples by GRADMM

#### C.3.1. POSITIVE LABEL

- I just got a new, high-end phone. It's got a new
- Joyful and sharing a good time with my friends. Life is so much better now
- Joy is a feeling that makes you feel like you are on the up and up
- I just got a new job working at a local Use of
- I am thrilled to be a new member of the twitterhub.

#### C.3.2. NEGATIVE LABEL

- I am so sad today. Sad is the word that I would use to write this
- I just received some sad and life news. I can't believe it. I am so
- I am so over it. I can't even believe it's over. I can't
- I just received news that my best-loved, and most-licked-at
- I just got back from a long day at work. I can't help

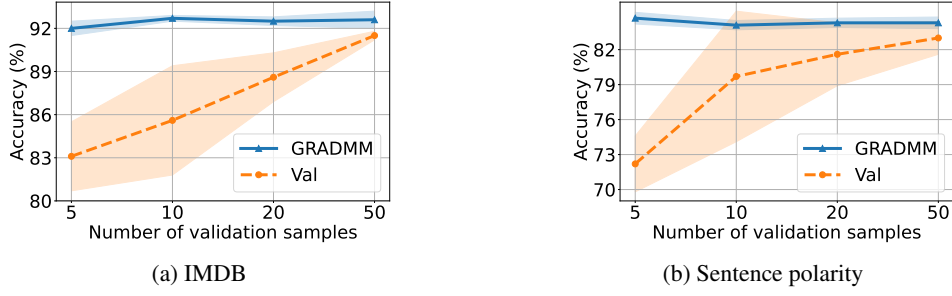


Figure 7. **Data-scarce regime.** Generating 100 synthetic samples with GRADMM, based on 5, 10, 20, 50 examples from a target task. Synthetic data generated based on only 5 real examples outperforms the real data by 8.9% and 12.5% on the two datasets.

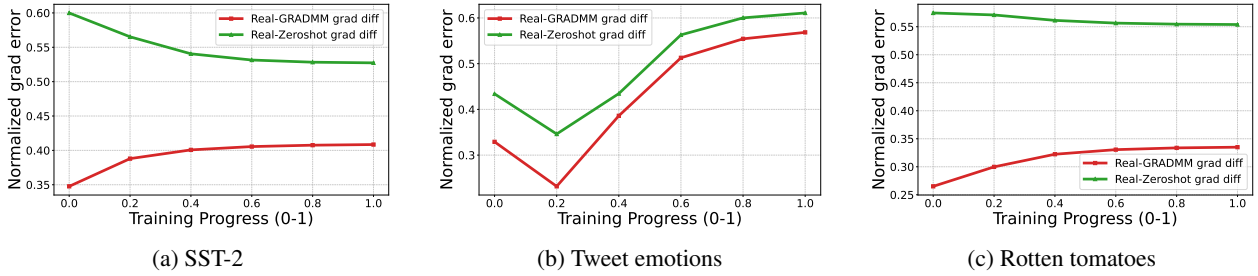


Figure 8. **Last-layer gradient differences** during fine-tuning on synthetic vs. real data. Data generated by GRADMM yields significantly smaller gradient errors compared to the zero-shot baseline, indicating closer alignment with real data.

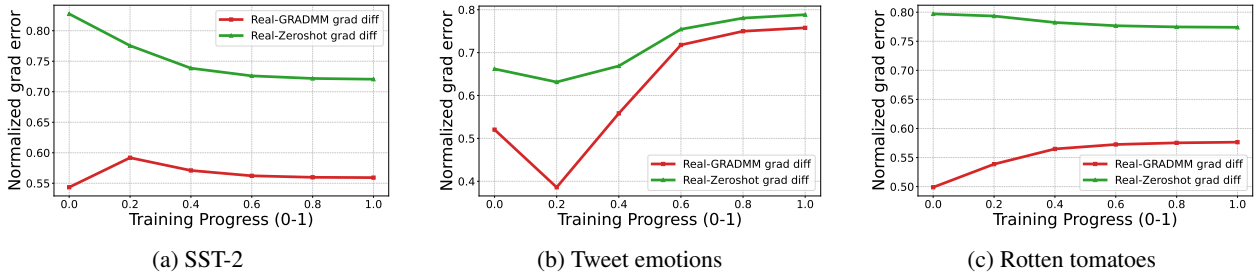


Figure 9. **Full gradient differences** during fine-tuning on synthetic vs. real data. Data generated by GRADMM yields significantly smaller gradient errors compared to the zero-shot baseline, indicating closer alignment with real data.

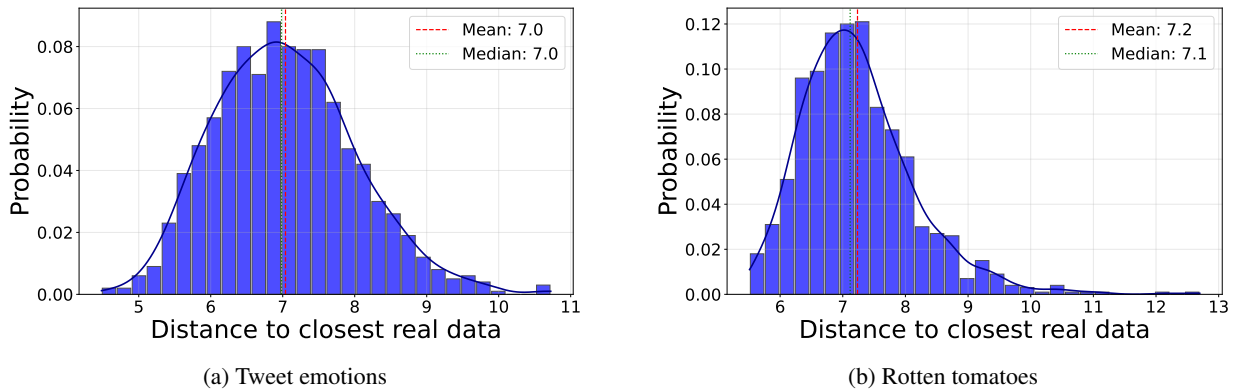


Figure 10.  $L_2$  embedding distance between GRADMM's synthetic texts to their closest real training data in (left) Tweet emotions and (right) Rotten tomatoes.

## D. Additional experiments

**Additional datasets.** We include two additional datasets namely IMDB (Maas et al., 2011) and Sentence polarity (Pang & Lee, 2005a) in the setting of Figure 1 in Section 5.2.1. Figure 7 shows the result of applying GRADMM to generate 100 synthetic examples based on only 5, 10, 20, 50 examples randomly selected from the validation data of IMDB and Sentence polarity. We see that GRADMM successfully generates high-quality supervised fine-tuning data that can train Phi to a superior performance over that of training on the available validation data. Notably, GRADMM generated synthetic data based on only 5 real examples outperform the real data by 8.9% and 12.5% on the two datasets. This confirms the effectiveness of GRADMM in the data-scarce regime.

**Last-layer gradient error.** Figure 8 demonstrates the normalized **last-layer** gradient error with respect to real data, i.e.  $(\|\nabla_{\theta_L} \mathcal{L}(\theta_t) - \nabla_{\theta_L} \mathcal{L}^s(\theta_t)\|) / \|\nabla_{\theta_L} \mathcal{L}(\theta_t)\|$ , remains low at the pretrained parameters at the pretrained parameters and continues to stay low throughout fine-tuning. Notably, the data generated by GRADMM yields a significantly lower gradient error than the zero-shot baseline during fine-tuning, supporting its better performance.

**Full gradient error.** Figure 9 shows that the normalized **full** gradient error with respect to real data, i.e.  $(\|\nabla_{\theta} \mathcal{L}(\theta_t) - \nabla_{\theta} \mathcal{L}^s(\theta_t)\|) / \|\nabla_{\theta} \mathcal{L}(\theta_t)\|$ . While GRADMM only matches the last-layer gradients of real and synthetic samples, we observe the same trend as that of last-layer gradient error in Figure 8. This reinforces our last-layer argument in Section 4.3.

**Embedding distance.** Figure 10 presents a histogram of the distances between synthetic examples and their nearest real training samples. The absence of synthetic examples that are extremely close to real data indicates that GRADMM does not simply replicate real training instances.